Computer Networking

Chapter 3: Transport Layer

Questions & Answers

Question 1

Connection-Oriented vs Connectionless Transport
Compare UDP (User Datagram Protocol) and TCP (Transmission Control Protocol) with respect to:

1. Reliability
2. Connection establishment
3. Flow control and congestion control
4. Suitable applications

Answer

(a) Reliability - **TCP:** Reliable, ensures in-order delivery using acknowledgments, sequence numbers, and retransmissions. - **UDP:** Unreliable, no error recovery or retransmissions.

(b) Connection Establishment - **TCP:** Requires a **3-way handshake** before data transfer (connection-oriented). - **UDP:** No handshake; data can be sent immediately (connectionless).

(c) Flow & Congestion Control - **TCP:** Implements both flow control (sliding window) and congestion control (AIMD, slow start). - **UDP:** None, sender transmits at chosen rate.

(d) Suitable Applications - **TCP:** Web browsing (HTTP/HTTPS), email (SMTP/IMAP), file transfer (FTP). - **UDP:** Real-time apps (VoIP, video streaming, online gaming) where low latency is preferred over reliability.

Final Note: TCP is reliable but heavier, UDP is lightweight but unreliable. Choice depends on application requirements.

Question 2

Explain the concept of **Reliable Data Transfer Protocols (rdt)**. Why are sequence numbers and timers important in protocols like rdt2.1, rdt2.2, and rdt3.0?

Answer

Reliable Data Transfer (RDT): The goal of RDT is to provide communication that is *correct, complete, and in-order*, even when the underlying channel may introduce errors or losses.

1. Sequence Numbers

- Distinguish between **new packets** and **retransmissions**.
- Prevent the receiver from accepting the same packet more than once.
- Essential when ACK/NAK messages are lost or corrupted.

2. Timers

- Ensure recovery when a packet or ACK is lost.
- Sender starts a timer for each transmitted packet.
- If no ACK is received before the timeout $\Rightarrow$ retransmit the packet.

3. Role in Specific Protocols

- **rdt2.1:** Uses sequence numbers to differentiate retransmissions when ACK/NAK packets may be corrupted.
- **rdt2.2:** NAK-free protocol; sequence numbers alone ensure correctness.
- **rdt3.0:** Adds **timers** to handle packet *losses* (not just corruption).

Question 3

Write your next question here.

Answer

Write the solution here.